

Lecture Notes: ETL on Cloud

Course Overview

Key Definitions

- **Cloud Computing:** The delivery of computing resources such as compute, storage, networking and managed services over the internet on a pay-as-you-use model. Users do not provision or maintain physical hardware.
- **Cloud ETL:** Performing extraction, transformation and loading workflows using cloud-native tools, services and compute engines that scale automatically with workload demand.
- **Managed Spark Services:** Cloud providers' Spark offerings (Dataproc, EMR, HDInsight) that automate cluster provisioning, scaling, configuration and maintenance.

Why ETL is Moving to the Cloud

Traditional ETL solutions rely heavily on fixed infrastructure that becomes a bottleneck as data volume and velocity increase. Cloud platforms solve this by providing:

- Elastic compute that expands or shrinks with workload size.
- Globally distributed storage that scales without capacity planning.
- Managed services that reduce operational overhead.
- Deep integration with analytics, ML and orchestration tools.

By the end of this module, you will be able to:

- Understand the fundamentals of cloud computing and its relevance to ETL.
- Identify major cloud provider services used in data engineering workflows.
- Execute ETL workloads using managed Spark services across GCP, AWS and Azure.
- Optimise ETL performance through cloud-native scaling and monitoring tools.

Session 1: Introduction to Cloud Computing

What is Cloud Computing?

Cloud computing provides access to computing resources over the internet without the need to purchase, install or maintain physical infrastructure. Instead of running applications and storing data on local servers, organisations leverage globally distributed data centres owned by cloud providers.

Core characteristics include:

- **On-Demand Availability:** Resources such as compute or storage can be provisioned instantly.
- **Measured Usage:** You pay only for what you use.
- **Resource Pooling:** Multiple users share cloud infrastructure securely.
- **Broad Network Access:** Services can be accessed from anywhere with internet connectivity.

Cloud computing enables flexibility, rapid deployment and high reliability, which makes it ideal for data-intensive ETL workloads.

Why Cloud for Data Engineering?

Modern organisations generate massive amounts of data—text, logs, events, transactions, streams—at high velocity and in various formats. Cloud platforms address these needs through:

Scalability for the 3Vs of Data

- Large **volumes** processed via distributed compute.
- High **velocity** managed through autoscaling and stream ingestion tools.
- Diverse **variety** handled using storage that supports structured and unstructured data.

Cost Efficiency

- No upfront server investment.
- Scale down during low usage periods.
- Pay-as-you-go encourages experimentation without financial risk.

Distributed Processing for ETL

Cloud platforms support distributed engines like Spark, enabling parallel transformations such as joins, aggregations and data cleaning across many nodes simultaneously.

Seamless Integration

Cloud ETL pipelines connect easily with databases, warehouses, ML services, orchestration tools and monitoring systems—all in a single ecosystem.

Cloud Service Models

Cloud models define how responsibility is divided between the provider and the user.

Infrastructure as a Service (IaaS)

You manage OS, runtime, applications and data. Provider manages hardware and virtualisation.

- Useful when you need maximum customisation.
- Example ETL use case: running custom Spark or Hadoop clusters.

Platform as a Service (PaaS)

Provider manages infrastructure, OS and runtime. You manage only application code.

- Useful for deploying applications without infrastructure overhead.
- ETL use case: running data preparation applications on serverless environments.

Software as a Service (SaaS)

Fully managed applications delivered over the internet.

- Example: analytics dashboards, CRM tools.
- ETL use case: using SaaS BI tools that query transformed datasets.

Cloud Deployment Models

Public Cloud

Shared cloud infrastructure accessible over the internet.

- Highly cost-efficient.
- Suitable for scalable ETL workloads.

Private Cloud

Dedicated infrastructure for a single organisation.

- Higher control and security.
- Useful for regulated industries.

Hybrid Cloud

Combination of on-premises + cloud environments.

- Helpful when certain data must remain on-premises.

Multi-Cloud

Use of multiple cloud providers simultaneously.

- Reduces dependency on a single provider.

Core Cloud Concepts

Regions, Zones and Data Centres

- **Region:** Geographic area hosting cloud services.
- **Availability Zones (AZs):** Physically separate locations within a region, ensuring fault tolerance.
- **Data Centres:** The physical facilities hosting servers, storage and networking equipment.

Compute Options

- **Virtual Machines (VMs):** Full control of OS and environment.
- **Containers:** Package code and dependencies for portability.
- **Serverless Functions:** Execute code without managing servers; ideal for event-driven tasks.

Storage Types

- **Object Storage:** Durable, scalable storage for unstructured data (e.g., logs, images, raw datasets).
- **Block Storage:** Low-latency storage suitable for databases.
- **File Storage:** Shared file systems for applications requiring POSIX-like structure.

Networking Components

- **VPCs** isolate cloud networks.
- **Load Balancers** distribute traffic.
- **Security Groups & Firewalls** define access rules.
- **VPNs** connect on-premises and cloud securely.

Cloud Advantages for ETL

Cloud platforms support ETL workloads through:

Distributed Execution

Data transformations run in parallel across clusters, drastically reducing processing time.

Elastic Scaling

Clusters expand and shrink automatically based on job complexity or data size.

Managed Services

Infrastructure provisioning, software patching and performance monitoring are automated.

Integration with Data Ecosystems

ETL outputs can be easily loaded into warehouses (BigQuery, Redshift) or queried serverlessly (Athena).

Session 2: Introduction to Cloud Platforms

Major Cloud Providers

Cloud ETL pipelines typically rely on one of the following:

Amazon Web Services (AWS)

Largest market share, mature ecosystem, strong analytics and machine learning capabilities.

Microsoft Azure

Enterprise-centric, strong integration with Windows, Active Directory and enterprise solutions.

Google Cloud Platform (GCP)

Strong data and ML ecosystem, BigQuery, AI-first tooling.

All provide globally distributed infrastructure, similar core services and unique strengths.

Cloud Storage Architecture & Services

Object storage is the central component for data engineering workflows.

Amazon S3

Highly durable (11 9's), lifecycle policies, tiering.

Azure Blob Storage

Supports hot/cool/archive tiers for cost optimisation.

GCS (Google Cloud Storage)

Uniform access patterns, tight integration with BigQuery and Dataproc.

Why Object Storage for ETL?

- Decouples storage from compute.

- Ideal for distributed read/write patterns of Spark.
- Supports massive scalability and parallel ingestion.

Compute Models & Virtualisation

Cloud compute services enable on-demand provisioning of VMs with different CPU, RAM and storage configurations.

Common features across providers

- Auto-scaling (scale out/in based on utilisation).
- Multiple machine families (general-purpose, memory-optimised, compute-optimised).
- Seamless integration with storage and networking layers.

ETL pipelines often use VMs as underlying nodes in Spark clusters.

Cloud Databases

Cloud platforms offer a broad spectrum of database services:

Relational Databases

MySQL, PostgreSQL, SQL Server.

Used for transactional workloads and staging ETL outputs.

NoSQL Databases

DynamoDB, MongoDB, Cosmos DB.

Used for high-velocity ingestion or semi-structured data.

Data Warehouses

BigQuery, Redshift, Snowflake.

Serve analytical queries and reporting needs.

Managed vs Self-hosted Databases

- Managed: automated backups, patching, scaling.
- Self-hosted: full control, custom configuration.

ML Services Overview

Cloud ML services provide:

- Pre-configured notebook environments.
- Managed training clusters.
- Hyperparameter tuning.

- Model hosting endpoints.

Examples:

- AWS SageMaker
- Azure ML
- Google Vertex AI

These services integrate directly with storage and ETL pipelines for seamless workflows.

PySpark on Cloud

Running Spark jobs on the cloud eliminates cluster management complexity.

Examples:

- **Dataproc (GCP):** Serverless and cluster-based Spark.
- **EMR (AWS):** Scalable Spark with S3 integration.
- **Synapse / HDInsight (Azure):** Spark pools integrated with Azure Data Lake.

All support:

- Job submission through UI or CLI.
- Log and metrics monitoring.
- Auto-scaling policies.

Identity & Access Management (IAM)

IAM governs who can access which cloud resources.

Key Concepts

- **Authentication:** Verifying identity.
- **Authorisation:** Assigning permissions.
- **Principle of Least Privilege:** Users receive only the access required.
- **Roles/Policies:** Define actionable permissions for users and services.

IAM is essential for secure data pipelines and ETL operations.

Session 3: ETL on the Cloud

Managed Spark Services for ETL

Spark is widely used for ETL due to distributed in-memory processing.

Why Managed Spark?

- No need to configure Hadoop/Spark manually.
- Automatic provisioning and scaling.
- Integration with cloud-native storage.
- Detailed monitoring and log analytics.

Cloud managed services make Spark accessible without needing deep cluster operations expertise.

Google Cloud Dataproc

Dataproc supports both:

- **Cluster-based Spark** for long-running workloads.
- **Serverless Spark** for on-demand execution.

ETL Workflow on Dataproc

1. Upload script and datasets to GCS.
2. Configure cluster (machine type, workers, Spark version).
3. Submit PySpark job through console or CLI.
4. Write outputs to GCS or BigQuery.

Serverless mode automatically provisions compute only for job duration, reducing cost for intermittent workloads.

Monitoring, Scaling & Cost Control

Cloud ETL workloads depend on effective monitoring.

Monitoring Tools Provide:

- Logs for debugging failures.
- Metrics for CPU, memory, network and shuffle operations.
- Job timelines and execution breakdowns.
- Alerts for anomalies.

Scaling Strategies

- **Vertical Scaling:** Increase node size.

- **Horizontal Scaling:** Add more workers.
- **Auto Scaling:** Adjust cluster size based on active jobs.

Cloud autoscaling helps maintain performance for large workloads while optimising cost.

Ask an LLM

1. Compare cluster-based Spark and serverless Spark for ETL workloads.
2. How does object storage simplify ETL staging and transformation layers?
3. Explain one advantage and one limitation of EMR, Dataproc and HDInsight.
4. Describe how autoscaling helps optimise cost during ETL.
5. Why is monitoring essential when running distributed Spark jobs?

Topics Commonly Asked in Cloud ETL Interviews

1. Differences between traditional ETL and cloud-native ETL.
2. How object storage acts as a data lake.
3. Choosing between Dataproc, EMR and HDInsight.
4. Best practices for IAM in ETL pipelines.
5. Handling skew and large shuffles in Spark-based ETL.
6. Differences between serverless and cluster-based ETL execution.
7. Cost optimisation strategies for ETL workloads.
8. How PySpark interacts with cloud storage layers.
9. Monitoring metrics used during Spark jobs.
10. When to use managed warehouses vs data lakes for ETL outputs.